Applied Economics

040178-1 and 040178-2
Winter Term 2025/2026
Language: English
4 ECTS

University of Vienna

Instructor: Pello Aspuru
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Dates and Location

Course code 040178-1

- Wednesdays: 09:45–11:15, starting October 1, 2025.
- PC-Seminarraum 5, Oskar-Morgenstern-Platz 1, Floor: KG01, Room: U1.611

Course code 040178-2

- Thursdays: 16:45–18:15, starting October 2, 2025.
- PC-Seminarraum 5, Oskar-Morgenstern-Platz 1, Floor: KG01, Room: U1.611

Course Description

This is a course in which students will reinforce the tools learned in Introductory Econometrics (BA) by applying them to real-world data sets using econometric software. Prior knowledge at that level will be assumed throughout. The aim is for students to get hands-on experience analyzing observational data using the open-source programming language R and to acquire the tools to carry out empirical research projects independently.

Course Objectives

In this course you will learn to:

- Use econometric tools to produce empirical analyses.
- Interpret and critically evaluate existing empirical studies.
- Understand possibilities and limitations of empirical analyses.

Required Software

- Install R from `https://cran.r-project.org/`
- Install RStudio from `https://posit.co/download/rstudio-desktop/`

Course Schedule

This is a tentative and non-binding course schedule:

- Week 1 (Oct 1st / 2nd): Introduction
- Week 2 (Oct 8th / 9th): Multiple Linear Regression 1/2

→ Deregistration period ends

- Week 3 (Oct 15th / 16th): Multiple Linear Regression 2/2
- Week 4 (Oct 22nd / 23rd): Ordinary Least Squares (OLS)
- Week 5 (Oct 29th / 30th): Inference 1/2
- Week 6 (Nov 5th / 6th): Inference 2/2

→ Week 7 (November 12th / 13th): Midterm exam

- Week 8 (Nov 19th / 20th): Causality and Correlation
- Week 9 (Nov 26th / 27th): Instrumental Variables (IV)
- Week 10 (Dec 3rd / 4th): Panel Data

→ Dec 5th, 23:59: Problem set online.

- Week 11 (Dec 10th / 11th): Differences in Differences (DD)
- Week 12 (Dec 17th / 18th): Regression Discountinuity Design (RDD)

→ Dec 19th, 23:59: Problem set submission deadline.

- Week 13 (Jan 7th / 8th): TBD
- Week 14 (Jan 14th / 15th): TBD
- Week 15 (Jan 21st / 22nd): TBD

→ Jan 29th, 16:45-18:15: Final exam

Grading Policy

Class Participation	10%
Problem Set	25%
Midterm Exam	25%
Final Exam	40%

Problem Set

- The problem set can be submitted in teams of up to three students.
- To register your problem set team:
  1. Enter this Google Sheet.
  2. Select your weekday tab.
  3. Add the team member names.

Academic Integrity

Students are expected to uphold the highest standards of academic honesty. Plagiarism or cheating will result in disciplinary action in accordance with university policy.

Accessibility

Students requiring accommodations should contact the instructor and the university's accessibility services.

Useful References

1. **Oswald** F, Viers V, Villedieu P, Kennedy G (2020). Introduction to Econometrics with R. SciencesPo Department of Economics, Paris, France.
   available online: https://scpoecon.github.io/ScPoEconometrics/.

2. **Heiss, F. (2020).** Using R for Introductory Econometrics (2nd edition)
   available online: http://urfie.net/readOnline.html

3. **Hanck**, C., Arnold, M., Gerber A., and Schmelzer M. (2023). Introduction to Econometrics with R
   available online: https://www.econometrics-with-r.org/index.html